Tomas Bernardo Navas Andrade
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EDUCATION

University of British Columbia, Vancouver, BC
Bachelor of Applied Sciences, Engineering (in progress)
Colegio Menor San Francisco, Quito, Ecuador
Diploma Type US High School, AP program

- GPA: 3.75
- Highlighted coursework: AP English Lang, AP Calc AB, AP Seminar, AP Spanish Lang, Advanced Physics, Advanced Chemistry

COMMUNITY SERVICE

Laura Barahona Kindergarten reconstruction, Quito, Ecuador — January 2023 - June 2023

- We provided maintenance for the school, fixing and painting walls, and building a playground train, concluding with a gift exchange for the children

Umoja Foundation, Arusha, Tanzania — March 15-30 2024

- We delivered clothing and school supplies to a local school, spent time with the community, and built a locker room and pens for poultry and livestock

Canadian Association of Medical Teams Abroad, Quito, Ecuador — Feb 5-16 2023

- We helped translate conversations between Canadian doctors and Ecuadorian patients who only spoke Spanish, and carried medical devices

EXTRACURRICULAR ACTIVITIES AND WORK EXPERIENCE

Member/Delegate/Chair , COMMNU Model United Nations — September 2023 —June 2025

- Delegate and Chair at models schools from different parts of Quito and Ecuador ; First Mention of Honor, Security Council CAMINU XXX model; Best delegate, Human Rights Council SEKNU XIX. Elected as chair for the OAS and NATO committees in COMMNU XXIII and CAENU XX. Award for leadership and dedication from the school club.

Store Employee, Vida Digital technology retail and repair — February 2022 - March 2022

- I cleaned and prepared the shelves and products before opening, evaluated and carried out hardware and software product repairs, and acquired industry insights.

Intern, Ecuador National Court of Justice — March 5-30 2023

- I shadowed various hearings and justices, and assisted them with research and briefs on different subjects.

Keyboardist, Colegio Menor G-Block band — August 2024 - July 2025

- I attended rehearsals for performances representing our school at internal and external events. Best keyboardist award at Birreria Schools Rock Battle and 3rd place overall band, preselected for scholarship at UDLA interscholar band contest, award for most improved student in music at Colegio Menor, partial 4

DESIGN PROJECTS

Casino Slot Machine- Arduino

- I designed and programmed an Arduino-based slot machine integrating LEDs, a button, Piezo buzzer and python serial.
- Implemented game logic, randomized outcomes, and serial communication between the Arduino hardware and a Python-based PC interface.

Thermodynamic application - “Thermoguns” (Concept Design)

- I designed two thermodynamics based devices applying heat transfer and gas law principles into a concept video game weapon system where the tradeoffs of different thermodynamic processes become different classes.
- Compared different thermodynamic processes and used their efficiency, heat transfer, and pressure behavior to shape the design tradeoffs of each concept.

STM32 Embedded Motor Control System

- Developed STM32 HAL firmware using ADC, PWM, and EXTI interrupts to control a DC motor and LED from potentiometer input.
- Debugged timer configuration, interrupt logic, and PWM behavior to achieve reliable motor-speed control.
- Built and hardware-tested the system on an STM32 NUCLEO-F401RE.

STM32-Based EV Battery Management System + Telemetry Simulator

- Extended the STM32 motor-control system with battery-management logic using NORMAL, WARNING, FAULT, and SHUTDOWN operating states.
- Integrated a MOSFET-based motor driver with external power and flyback protection, alongside I2C temperature sensing and system-status indicators.
- Simulated battery telemetry including voltage, current, temperature, and state-of-charge data for monitoring and fault detection.
- Implemented threshold-based protection logic to detect unsafe operating conditions and trigger appropriate system responses.

SKILLS

Software: C, C++, Python, GitHub, STM32 Cube IDE/Cube MX, MATLAB, Simulink

Hardware: Arduino, STM32, KiCad, circuit prototyping, hardware/software debugging

Training: *Embedded Systems Development with C* — Coursera (2026); 15+ Arduino projects. **In progress :** ARM get Practical with Hardware, Modeling and simulation with simulink (Coursera)

Languages: Spanish (Native), English (C1), French (C1)

Professional: Teamwork, communication, leadership, adaptability, emotional intelligence

Interests: Music & performance, psychology, electronics, embedded systems, engineering design